

**Министерство образования и науки Российской Федерации ФГАОУ ВО
«УрФУ имени первого Президента России Б.Н. Ельцина» Кафедра
«Институт радиоэлектроники и информационных технологий-РТФ»**

**ПРАКТИЧЕСКАЯ РАБОТА №7.
MULTICHAIN DAPPS**

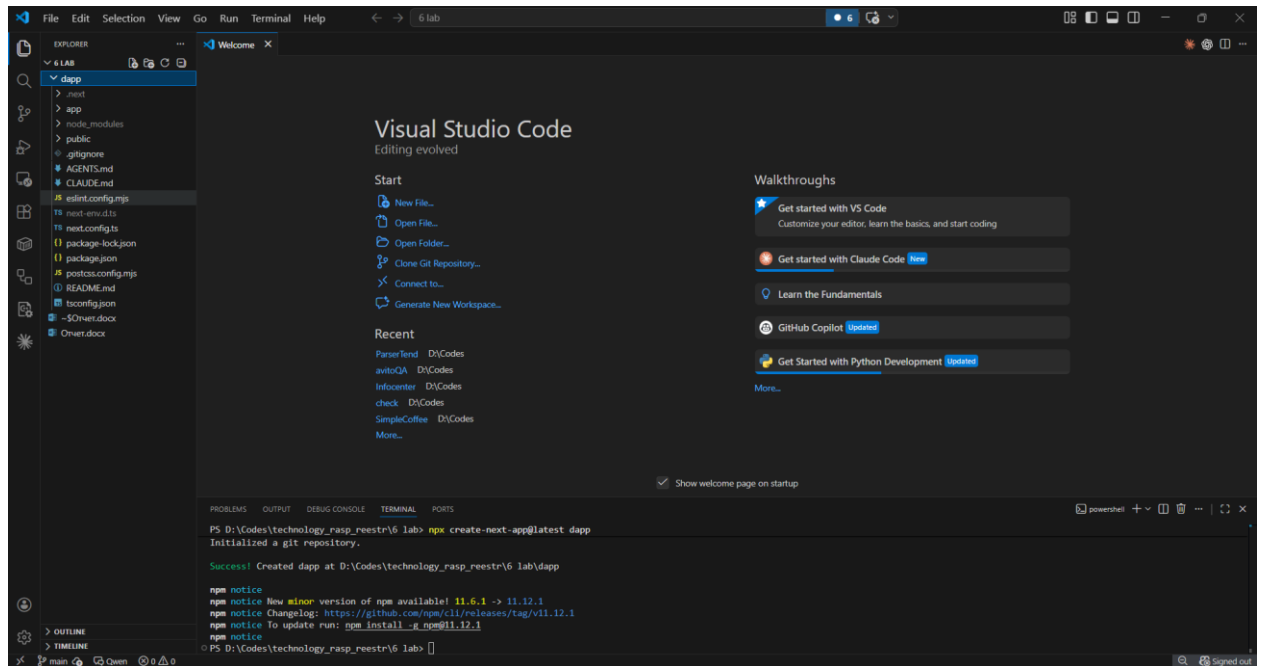
Преподаватель: Саиф Муджахед Абдулла Хаель

Студент: Кисляков Максим Вячеславович

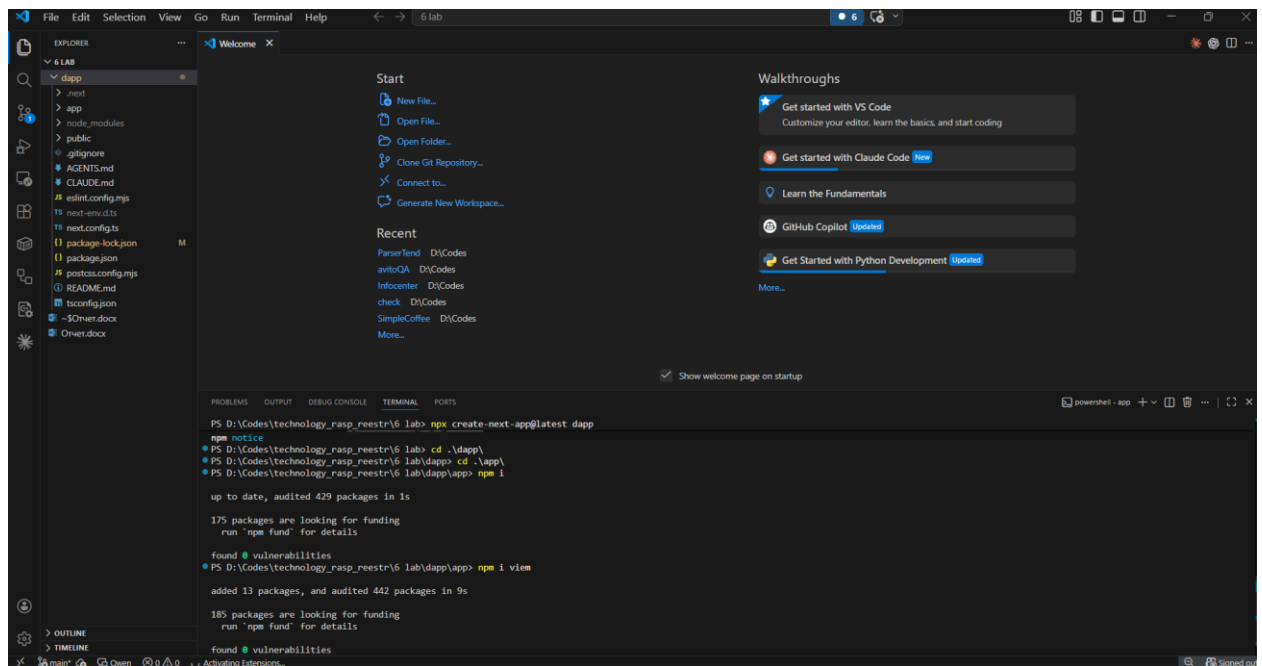
Группа: РИ-330936

Екатеринбург 2026

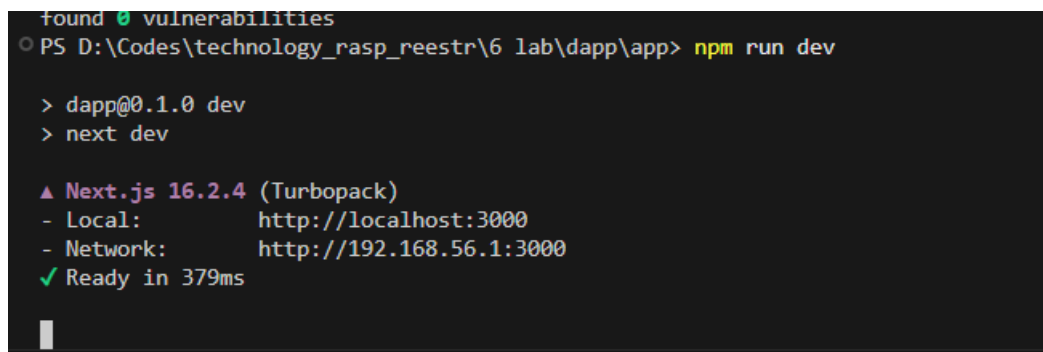
Создание нового проекта



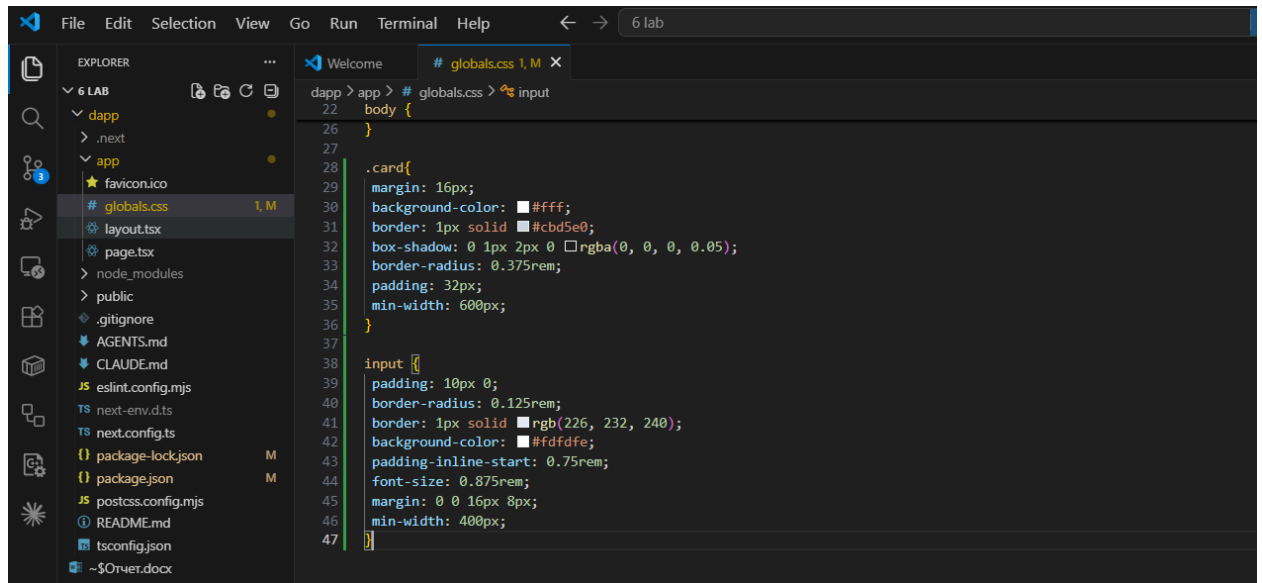
Установка зависимостей



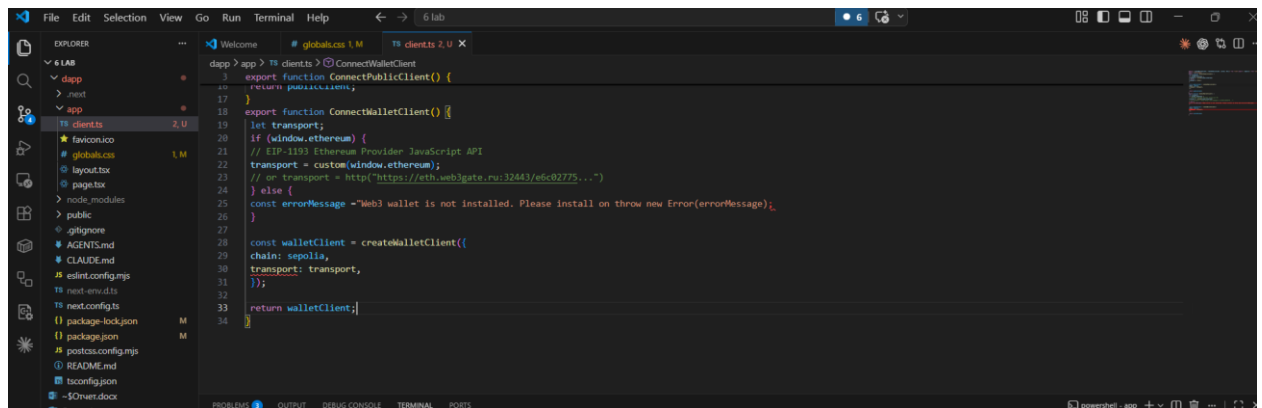
Приложение работает



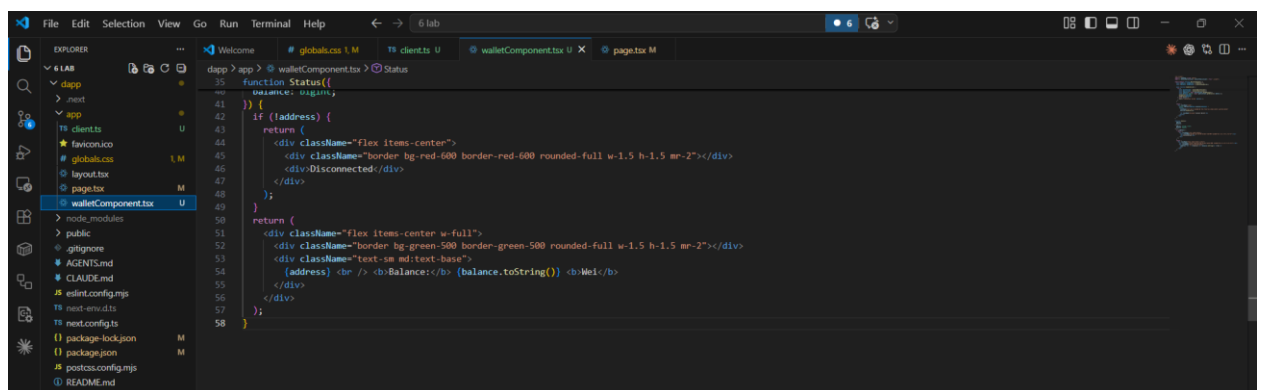
Подключение стилей



Подключение библиотеки viem

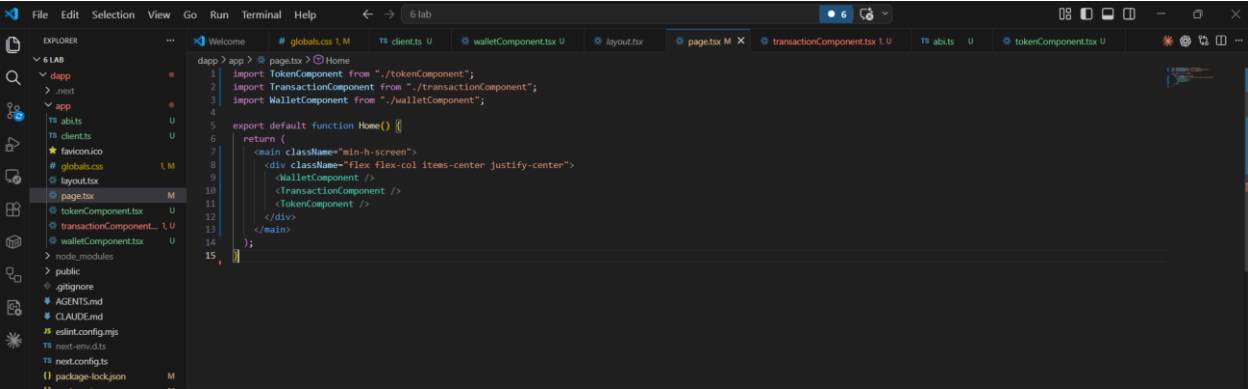


WalletComponent.tsx

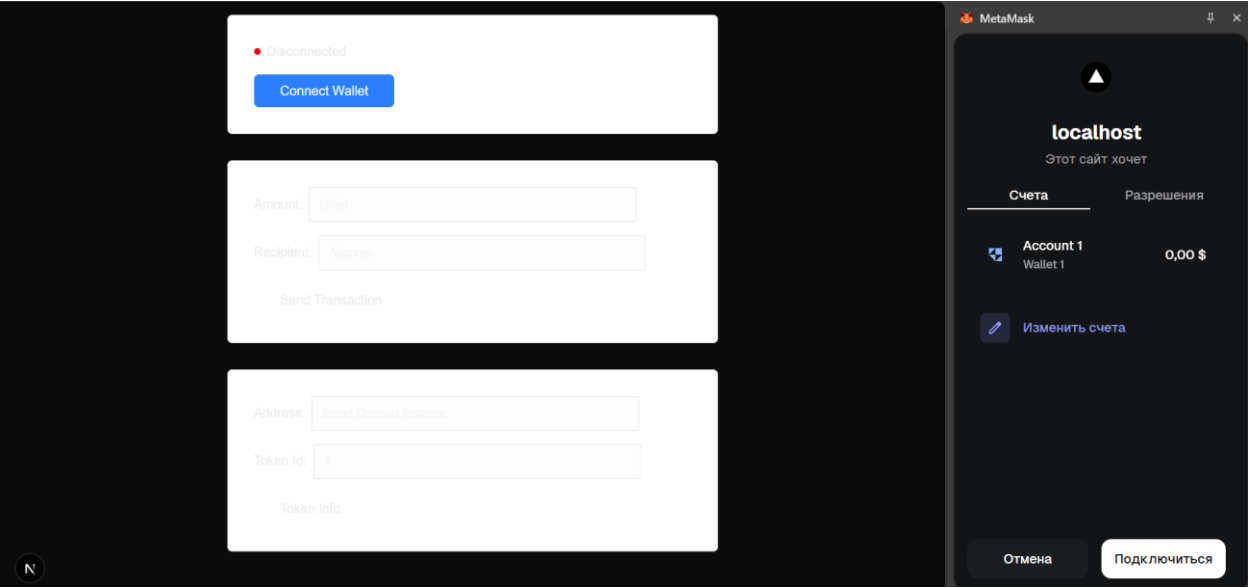


Page.tsx

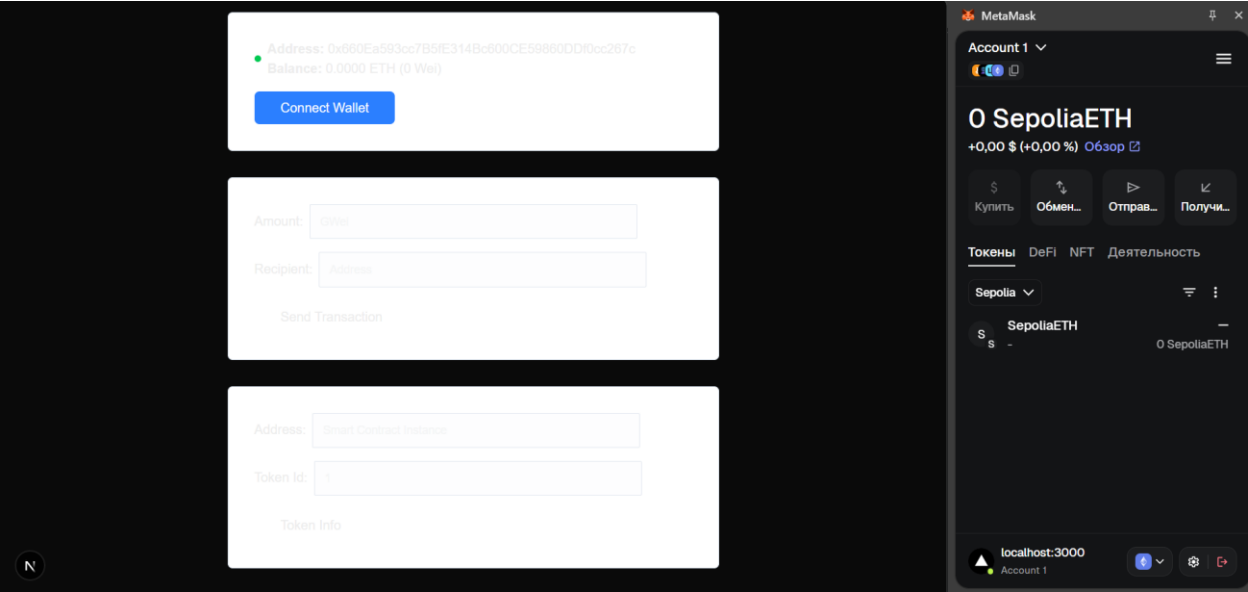
Обновление page.tsx



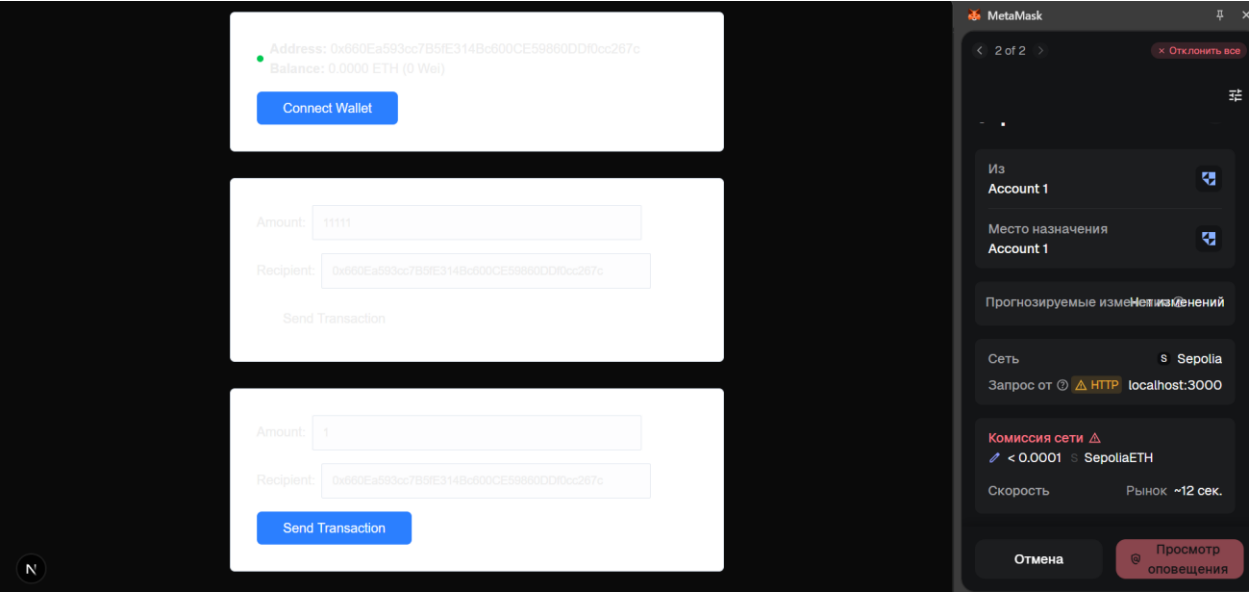
Проверка работоспособности



Подключение кошелька



Проверка отправки (не получается из за комиссии сети)

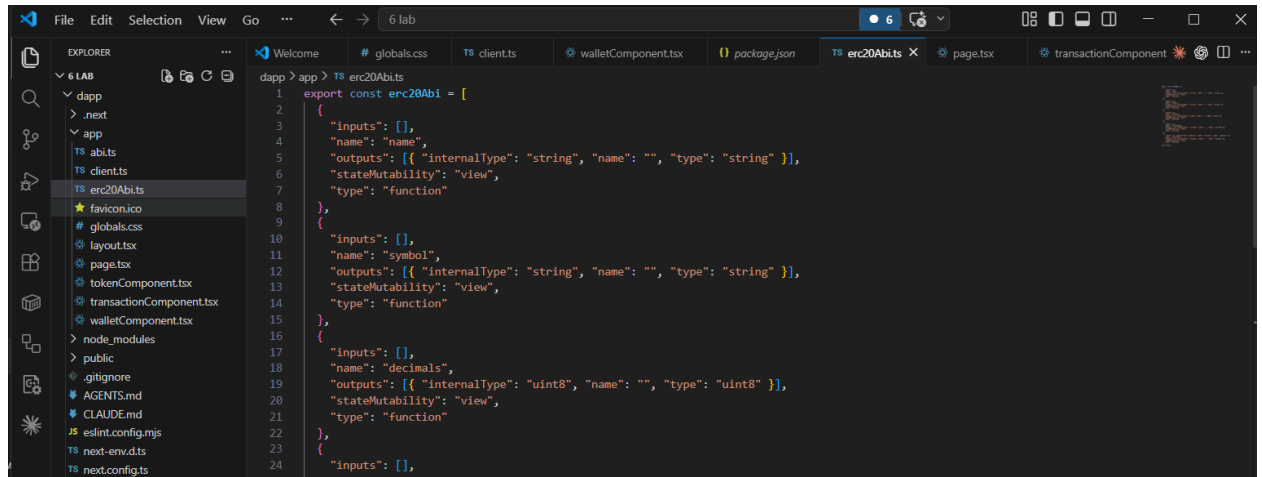


Самостоятельная работа

Беру стандартный контракт **ERC20** токена

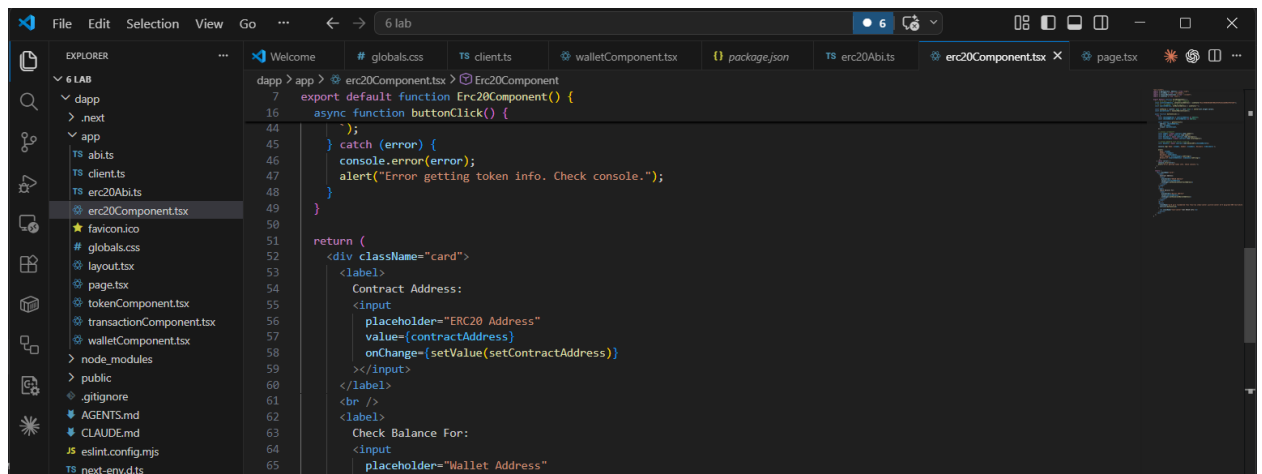
В качестве примера использую **USDC (Circle USD)** в тестовой сети Sepolia. Адрес контракта:
0x1c7D4B196Cb0C7B01d743Fbc6116a902379C7238

ABI стандартного токена



```
1 export const erc20Abi = [
2   {
3     "inputs": [],
4     "name": "name",
5     "outputs": [{ "internalType": "string", "name": "", "type": "string" }],
6     "stateMutability": "view",
7     "type": "function"
8   },
9   {
10    "inputs": [],
11    "name": "symbol",
12    "outputs": [{ "internalType": "string", "name": "", "type": "string" }],
13    "stateMutability": "view",
14    "type": "function"
15  },
16  {
17    "inputs": [],
18    "name": "decimals",
19    "outputs": [{ "internalType": "uint8", "name": "", "type": "uint8" }],
20    "stateMutability": "view",
21    "type": "function"
22  },
23  {
24    "inputs": [],
25    "name": "totalSupply",
26    "outputs": [{ "internalType": "uint256", "name": "", "type": "uint256" }],
27    "stateMutability": "view",
28    "type": "function"
29  },
30  {
31    "inputs": [{"name": "to", "type": "address"}, {"name": "value", "type": "uint256"}],
32    "name": "transfer",
33    "outputs": [{ "internalType": "bool", "name": "", "type": "bool" }],
34    "stateMutability": "nonpayable",
35    "type": "function"
36  },
37  {
38    "inputs": [{"name": "to", "type": "address"}, {"name": "value", "type": "uint256"}, {"name": "data", "type": "bytes"}],
39    "name": "transferFrom",
40    "outputs": [{ "internalType": "bool", "name": "", "type": "bool" }],
41    "stateMutability": "nonpayable",
42    "type": "function"
43  },
44  {
45    "inputs": [{"name": "spender", "type": "address"}, {"name": "value", "type": "uint256"}],
46    "name": "approve",
47    "outputs": [{ "internalType": "bool", "name": "", "type": "bool" }],
48    "stateMutability": "nonpayable",
49    "type": "function"
50  },
51  {
52    "inputs": [{"name": "owner", "type": "address"}, {"name": "spender", "type": "address"}],
53    "name": "allowance",
54    "outputs": [{ "internalType": "uint256", "name": "", "type": "uint256" }],
55    "stateMutability": "view",
56    "type": "function"
57  }
58 ]
```

app/erc20Component.tsx



```
1 import { useState } from 'react';
2 import { ethers } from 'ethers';
3 import { erc20Abi } from '../abi/erc20Abits';
4
5 export default function ERC20Component() {
6   const [contractAddress, setContractAddress] = useState('');
7   const [walletAddress, setWalletAddress] = useState('');
8   const [balance, setBalance] = useState('');
9
10  async function handleClick() {
11    if (!contractAddress || !walletAddress) {
12      console.error('Please enter both contract and wallet addresses');
13      return;
14    }
15    try {
16      const provider = new ethers.providers.Web3Provider(window.ethereum);
17      const contract = new ethers.Contract(contractAddress, erc20Abi, provider);
18      const balance = await contract.balanceOf(walletAddress);
19      setBalance(ethers.utils.formatEther(balance));
20    } catch (error) {
21      console.error(error);
22      alert('Error getting token info. Check console.');
23    }
24  }
25
26  return (
27    <div className="card">
28      <label>
29        Contract Address:
30        <input
31          type="text"
32          placeholder="ERC20 Address"
33          value={contractAddress}
34          onChange={setContractAddress}
35        />
36      </label>
37      <br />
38      <label>
39        Check Balance For:
40        <input
41          type="text"
42          placeholder="Wallet Address"
43          value={walletAddress}
44          onChange={setWalletAddress}
45        />
46      </label>
47      <button onClick={handleClick}>Get Balance</button>
48    </div>
49  );
50 }
```

Добавление на страницу нового элемента

localhost:3000

Amount:

Recipient:

Send Transaction

Amount:

Recipient:

Send Transaction

Contract Address:

Check Balance For:

Get ERC20 Info

Проверка работоспособности

Подтвердите действие на localhost:3000

Name: USDC
Symbol: USDC
Decimals: 6
Total Supply: 10883406005783602370
Balance of 0x660Ea593cc7B5fE314Bc600CE59860DDf0cc267c: 0

OK

Amount:

Recipient:

Send Transaction

Contract Address:

Check Balance For:

Get ERC20 Info

Ссылка на GitHub:

https://github.com/MaksimKislyakov/technology_rasp_reestr/tree/main/6%20lab